

MTH203: Multivariate Calculus

Monsoon 2025

Name: _____

Roll No.: _____

Tutorial Quiz 8

Question:

1. Evaluate $\int_C xy \, dx + (x + y) \, dy$ along the curve $y = x^2$ from $(-1, 1)$ to $(2, 4)$. (1)
2. Find the work done by $\mathbf{F} = (x^2 + y)\mathbf{i} + (y^2 + x)\mathbf{j} + ze^z\mathbf{k}$ over the following paths from $(1, 0, 0)$ to $(1, 0, 1)$:
 - (a) The line segment $x = 1, y = 0, 0 \leq z \leq 1$
 - (b) The x-axis from $(1, 0, 0)$ to $(0, 0, 0)$ followed by the parabola $z = x^2, y = 0$ from $(0, 0, 0)$ to $(1, 0, 1)$

(2)

Please note: There is no fractional marking. Only integer marking.